PRESCRIBING FINITELY MANY VALUES OF A SECTION AT PAIRWISE DISTINCT MAXIMAL IDEALS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Chinese remainder: for pairwise distinct maximal ideals the product map M -> the direct sum of the quotients M(x_i) is surjective. invoked at proof lines 38802-38806. Provides the surjection M -> M(x_1) plus ... plus M(x_n) whose preimages give the required section.

Source reference. Original lemma 00DT: sources/stacks-chow/tags.txt, source lines 509–509. Cited at proof lines 38802-38806 for the surjection M -> M(x_1) plus ... plus M(x_n); the surjectivity is load-bearing inside the counted span.

Source reference. Original definition 0GV8: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38721–38721. Cited at statement lines 38795-38796 for the setup the lemma quantifies over; the notation (M, x_i, V(x_i)) is load-bearing inside the counted span.

Lemma 0.2. *In Situation 0GV8 let $x_1, \ldots, x_n \in \Omega$ be pairwise distinct. Let $v_i \in V(x_i)$. Then there exists an $s \in M$ such that $s(x_i)$ maps to v_i for $i = 1, \ldots, n$.*

Proof. Since x_i is a maximal ideal of R we may use Algebra, Lemma 00DT to see that $M(x_1) \oplus \ldots \oplus M(x_n)$ is a quotient of M. □

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.